MATH73235 Spring 2025 Quiz 2

First name:

Last name:

Student ID:

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Problem 1. *Find $\vec{\nabla} f$ and its value at P for*

$$f = \frac{x}{x^2+y^2}, \quad P(1,1).$$

Solution.

$$\vec{\nabla} f = \left[\frac{\partial}{\partial x} f, \frac{\partial}{\partial y} f\right]$$

$$\frac{\partial}{\partial x} f = \frac{\partial}{\partial x}\left(\frac{x}{x^2+y^2}\right) = \frac{(x^2+y^2)\frac{\partial}{\partial x}x - x\frac{\partial}{\partial x}(x^2+y^2)}{(x^2+y^2)^2} \quad \text{[Quotient Rule]}$$

$$= \frac{(x^2+y^2)(1) - x(2x)}{(x^2+y^2)^2}$$

$$= \frac{x^2+y^2-2x^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\frac{x}{(x^2+y^2)} = x\frac{\partial}{\partial x}(x^2+y^2)^{-1} = x(-1)(x^2+y^2)^{-2}\frac{\partial}{\partial y}(y^2) \quad \text{[Chain Rule]}$$

$$= \frac{-x}{(x^2+y^2)^2}(2y) = \frac{-2xy}{(x^2+y^2)^2}$$

So $\vec{\nabla} f = \begin{bmatrix} \frac{y^2-x^2}{(x^2+y^2)^2} \\ \frac{-2xy}{(x^2+y^2)^2} \end{bmatrix}$

at $P(1,1)$:

$$\vec{\nabla} f\Big|_{(1,1)} = \begin{bmatrix} \frac{1^2-1^2}{(1+1)^2} \\ \frac{-2(1)(1)}{(1^2+1^2)^2} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{-2}{2^2} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{1}{2} \end{bmatrix}$$

$\square$

What I am looking for:
- Correct differentiation
- Definition of $\vec{\nabla} f$ (stated or implied)
- Calculating $\vec{\nabla} f|_{(1,1)}$ correctly.

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Problem 2. *Find* div $\vec{v}$ *and its value at* P *for*

$$\vec{v} = \begin{bmatrix} 0 \\ \cos(xyz) \\ \sin(xyz) \end{bmatrix}, \quad P(2, \frac{\pi}{2}, 0).$$

Solution.

$$\vec{\nabla}\cdot\vec{v} = \frac{\partial}{\partial x} v_1 + \frac{\partial}{\partial y} v_2 + \frac{\partial}{\partial z} v_3 = \frac{\partial}{\partial x} 0 + \frac{\partial}{\partial y} \cos(xyz) + \frac{\partial}{\partial z} \sin(xyz)$$

$$= 0 - \sin(xyz)(xz) + \cos(xyz)(xy) \quad [\text{Chain Rule}]$$

$$= xy\cos(xyz) - xz\sin(xyz)$$

Then $$\vec{\nabla}\cdot\vec{v}\Big|_{(2, \pi/2, 0)} = (2)(\pi/2)\cos\left(2\cdot\frac{\pi}{2}\cdot 0\right) - (2)(0)\sin\left(2\cdot\pi/2\cdot 0\right)$$

$$= \pi\cos(0) - 0\sin(0)$$

$$= \pi$$

What I'm looking for:

- definition of $\vec{\nabla}\cdot\vec{v}$ (stated or implied)
- Correct calculation of $\vec{\nabla}\cdot\vec{v}$ using Chain Rule
- Value at P.

$\square$